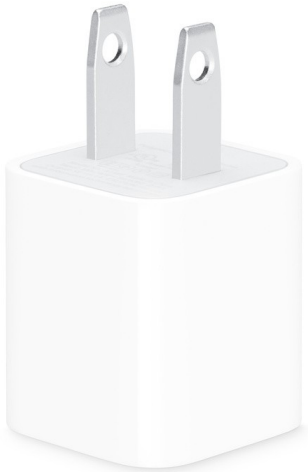
Master’s thesis defense · DTU Electro

Design and Implementation of a High-Frequency Charge-Pump PFC Resonant Converter for a USB-C Power Supply

Richard Xu

USB adapters: power range and compact design



Apple 5 W USB power adapter



Apple 20 W USB-C power adapter

Apple 140 W USB-C power adapter

Project target: 100 W continuous

Minimum: 65 W continuous

Higher switching frequency brings a loss tradeoff

Higher switching frequency

Smaller passive components

More switching loss and heat

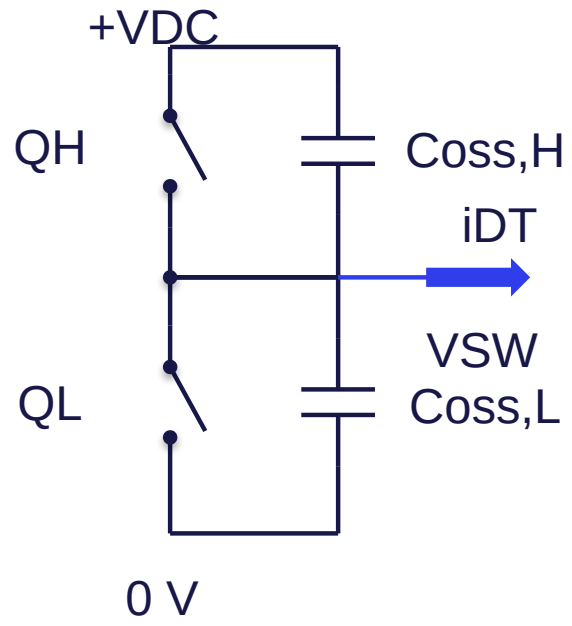
$$P_{sw} \approx f_s (E_{on} + E_{off})$$

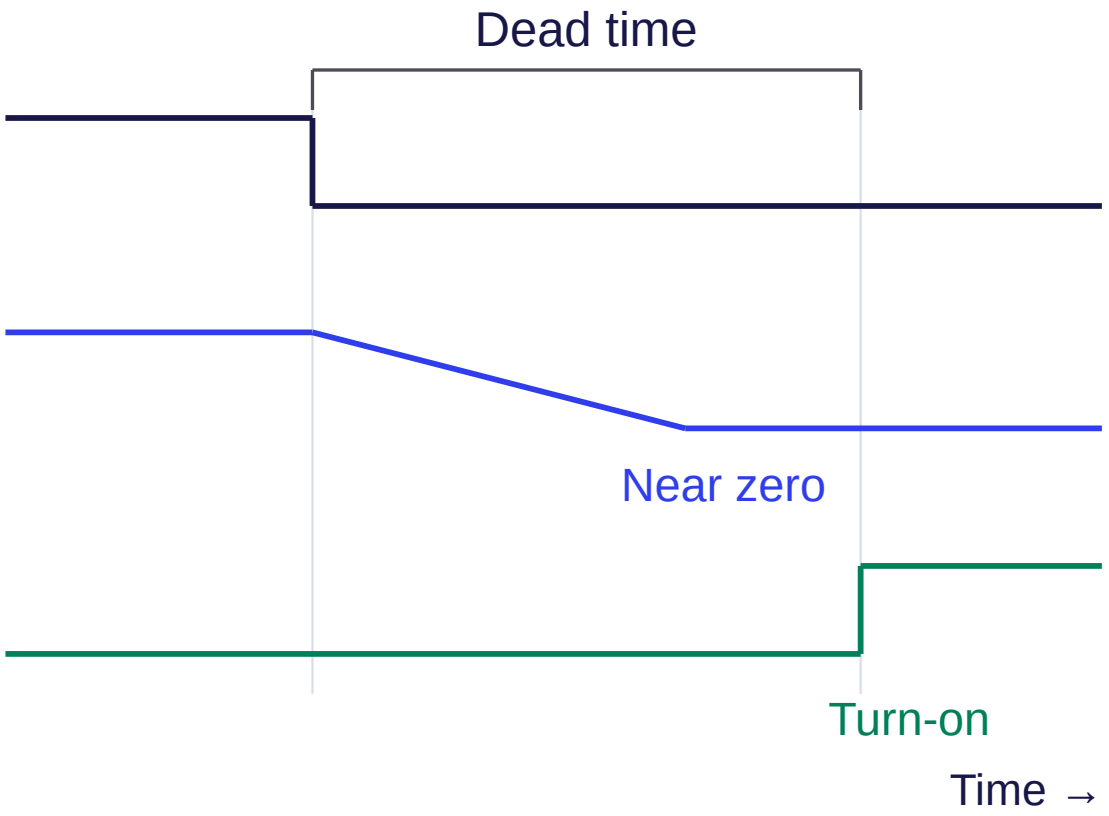
For fixed energy per transition

Conceptual tradeoff

ZVS reduces turn-on loss

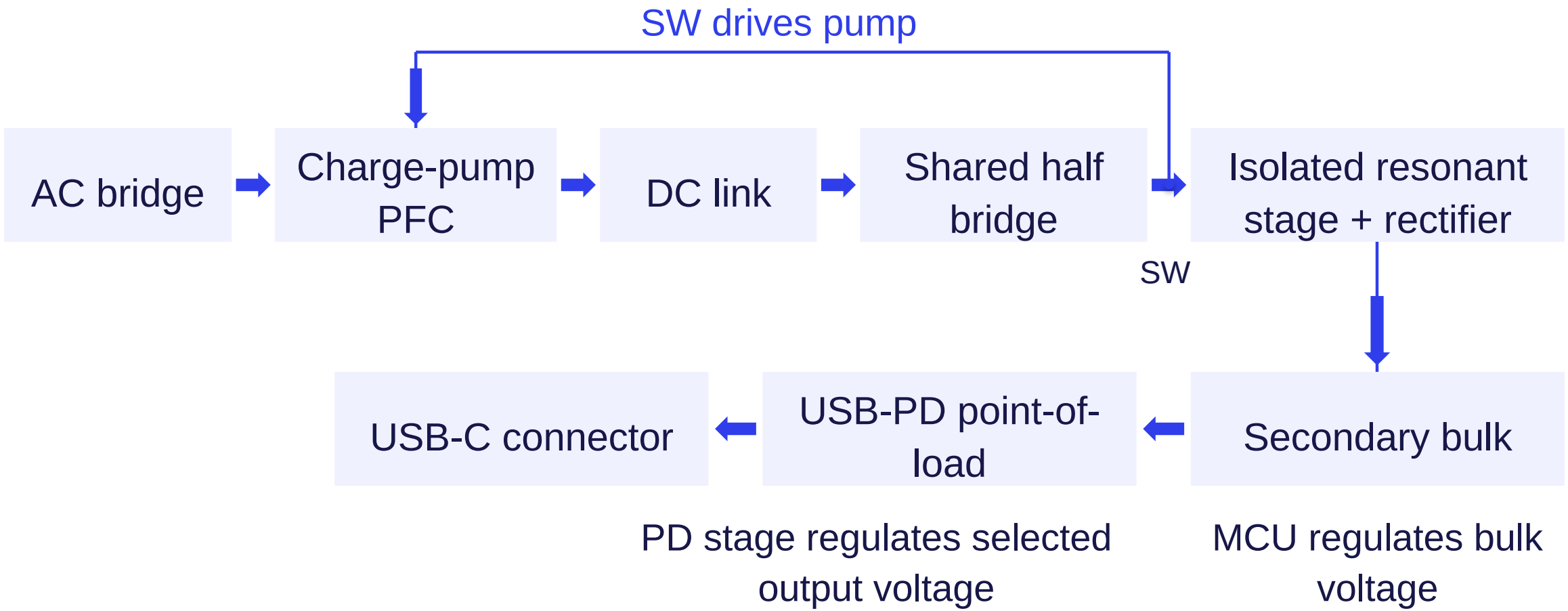
Low-side turn-on · conceptual





- Circuit current commutates capacitances during dead time
- Current and timing vary with operating condition

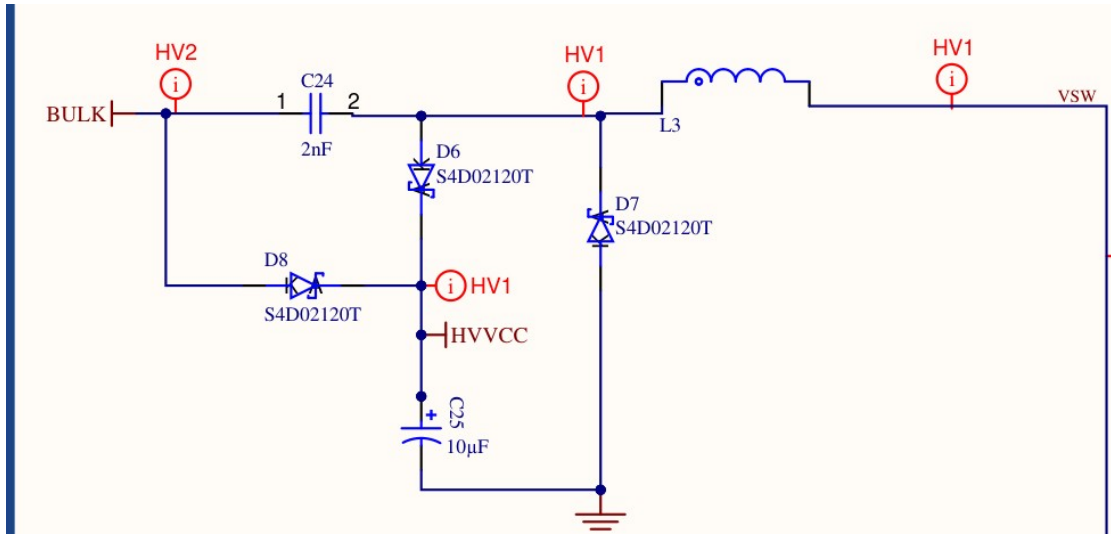
Charge-pump PFC feeds an isolated USB-C supply



Topology and magnetics: Ammar et al., 2020, 2021

Charge transfer shapes the input current

V2 charge-pump circuit



C24: pump capacitor C_p

L3: pump inductor L_p

VSW: shared half-bridge node

$$\Delta Q \approx C_p v_{rec}$$

$$\bar{I}_{rec} \approx b\, f_s\, C_p\, v_{rec}$$

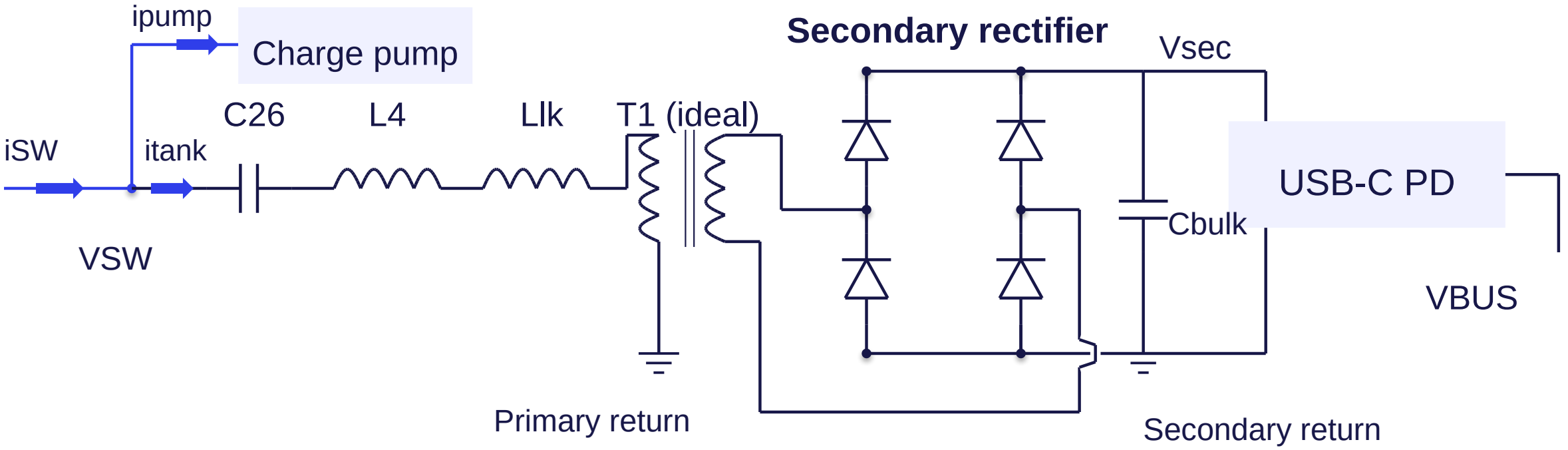
Ideal complete excursion; b = active-cycle density

- Average current follows rectified line voltage.
- Burst density controls average charge transfer.
- No separate dedicated PFC current loop.

Ammar et al., 2021

The resonant path supplies the isolated bulk voltage

V2 resonant path · simplified primary-referred equivalent



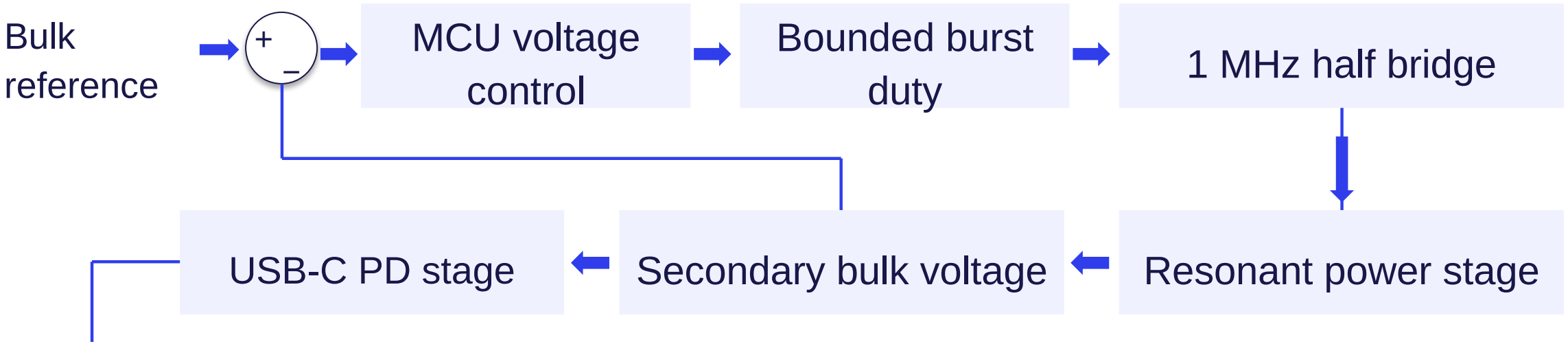
iSW = ipump + itank

Llk: transformer leakage

MCU: bulk regulation

PD: selected USB-C voltage

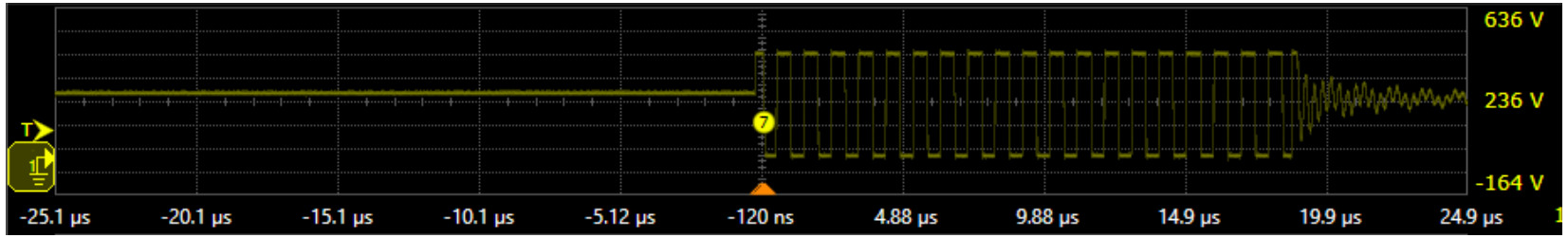
Secondary-voltage feedback sets burst duty



- Loaded duty is trimmed from bulk-voltage error.
- Startup uses bounded energy packets; no-load duty is capped.
- Overvoltage reduces or inhibits transfer.

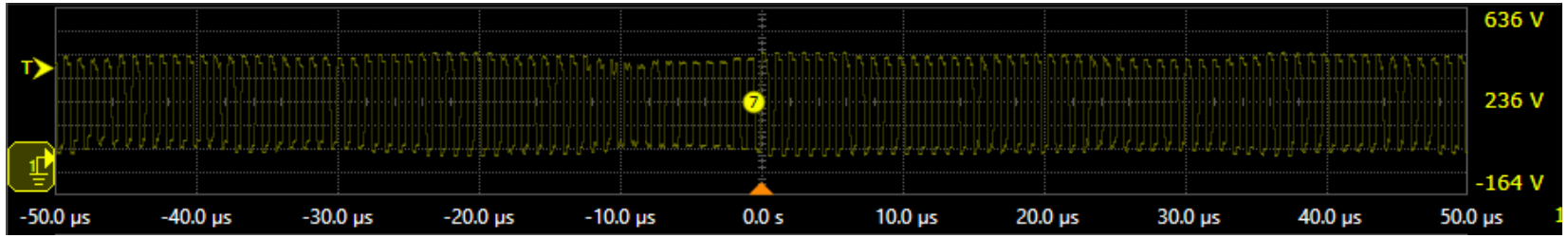
V1 switching changes with the operating point

5 V / 3 A · 5 µs/div



Idle Switching packet

20 V / 4 A · 10 µs/div

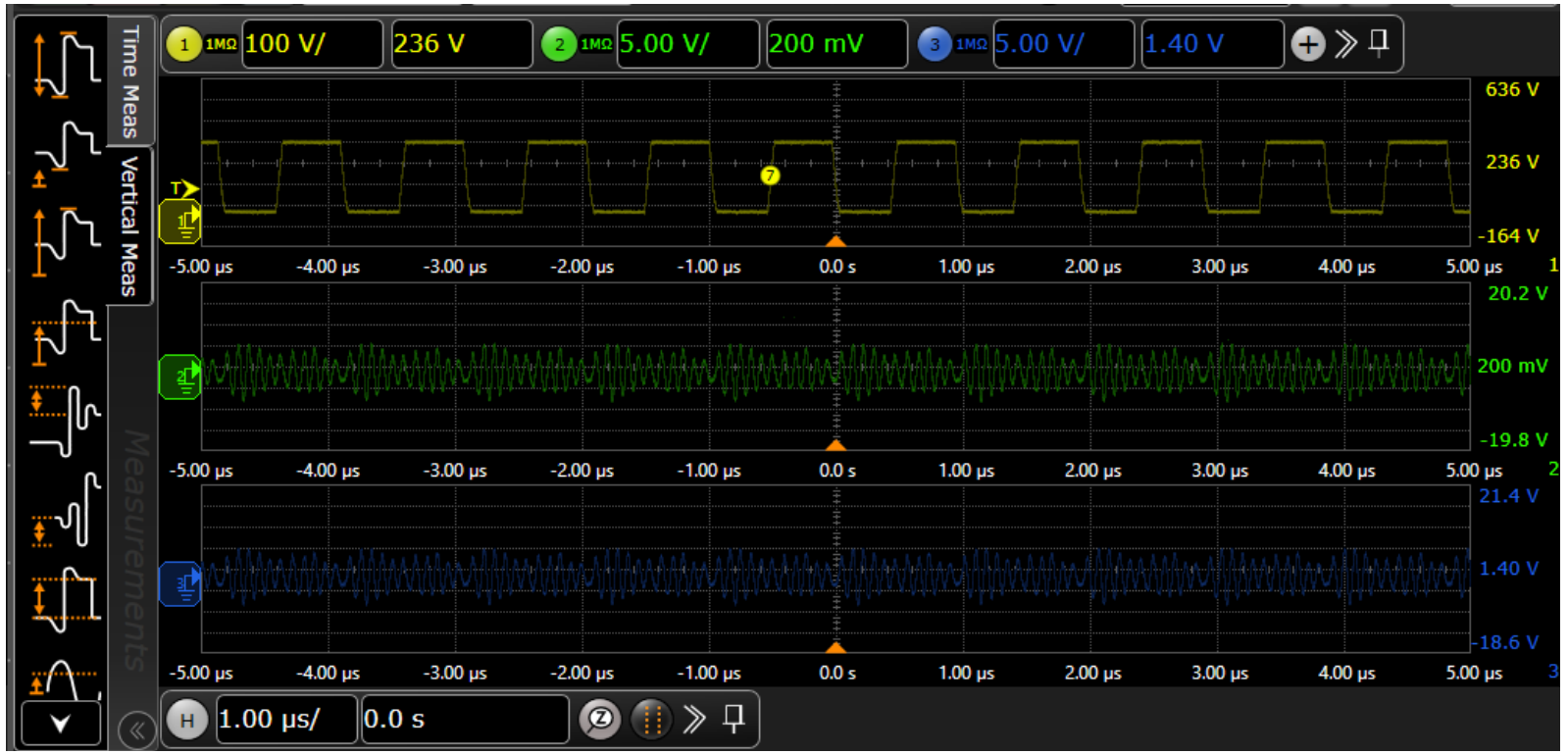


- 5 V / 3 A: switching packet and idle interval
- 20 V / 4 A: switching across the displayed 100 µs
- Active cycles near 1 MHz

Yellow: half-bridge switch-node voltage · 100 V/div

V1 switching at approximately 1 MHz

20 V USB-C setting, 4.7 A load



Yellow: switch node · Green: secondary output · Blue: USB-C VBUS

Gate timing is needed to assess ZVS

Secondary and VBUS channels AC-coupled; 1 µs/div

V1 delivered 79.89 W at USB-C

79.89 W at USB-C

19.9732 V × 4.0 A = 79.89 W

V1 · 240 V AC, 50 Hz input · electronic load

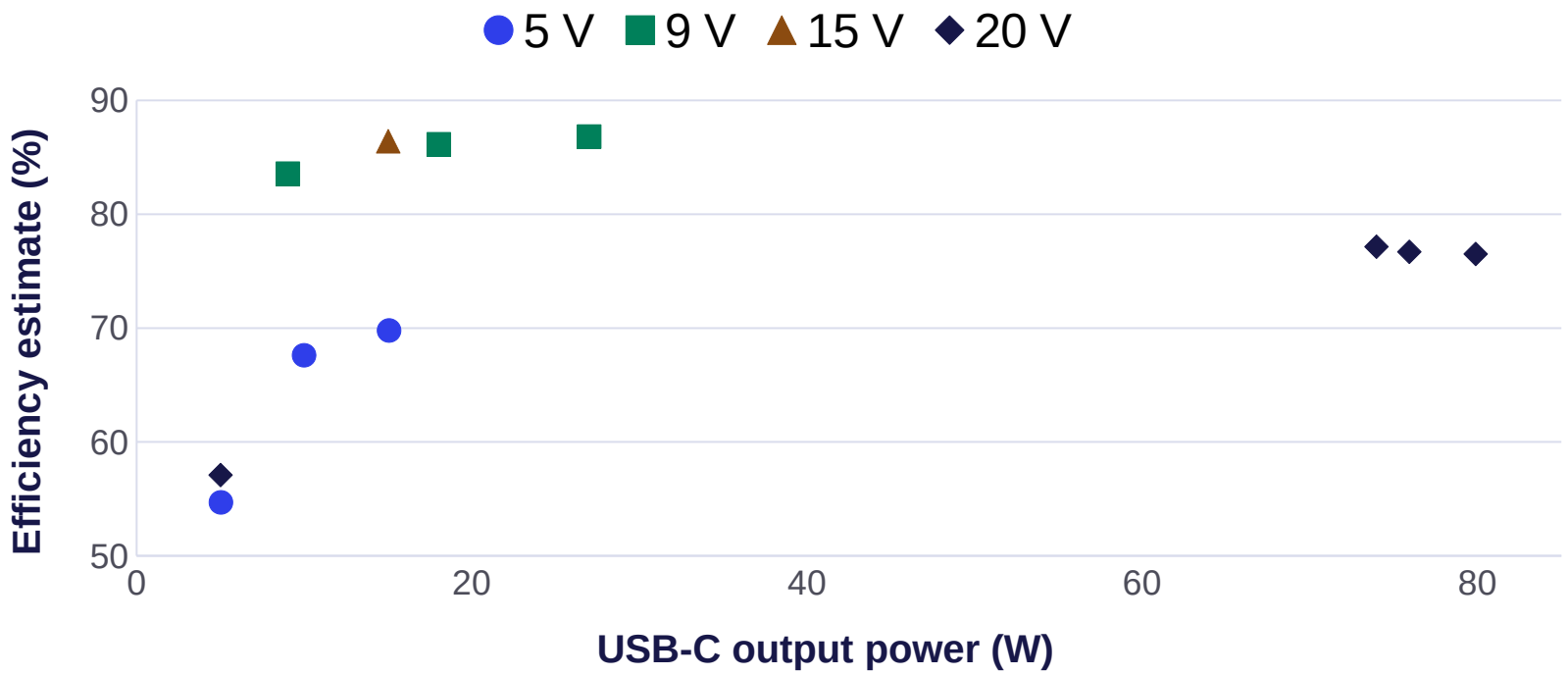
Selected voltage (V)	Load setting (A)	Mean VBUS (V)
5	3.0	5.02118
9	3.0	9.00057
15	3.0	14.98350
20	4.0	19.97320

Selected loaded points at USB-C

- Largest average-voltage error across loaded points: 0.583%

Efficiency estimate was 76.51% at 79.89 W

V1 initial efficiency estimates



104.418 W input → 79.8928 W USB-C output

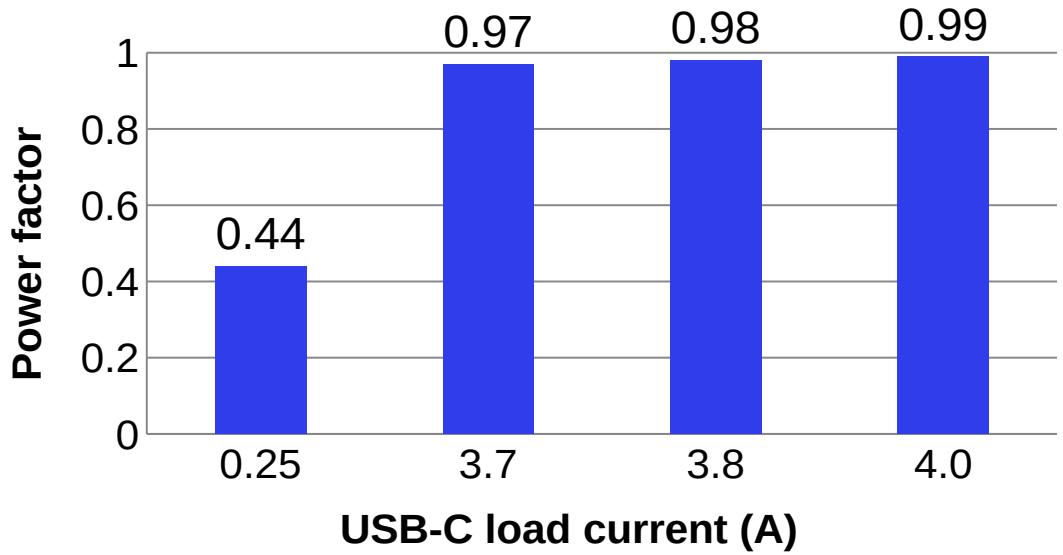
Input–output difference: 24.53 W

79.89 W, 76.51%

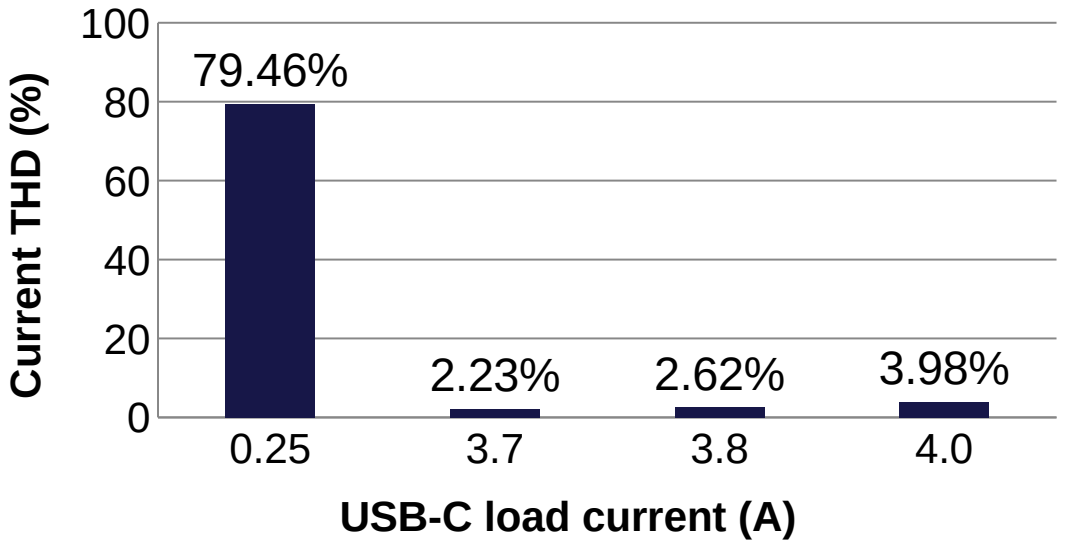
Input power quality improved at higher load

V1 · 20 V USB-C setting · 240 V AC, 50 Hz

Input power factor

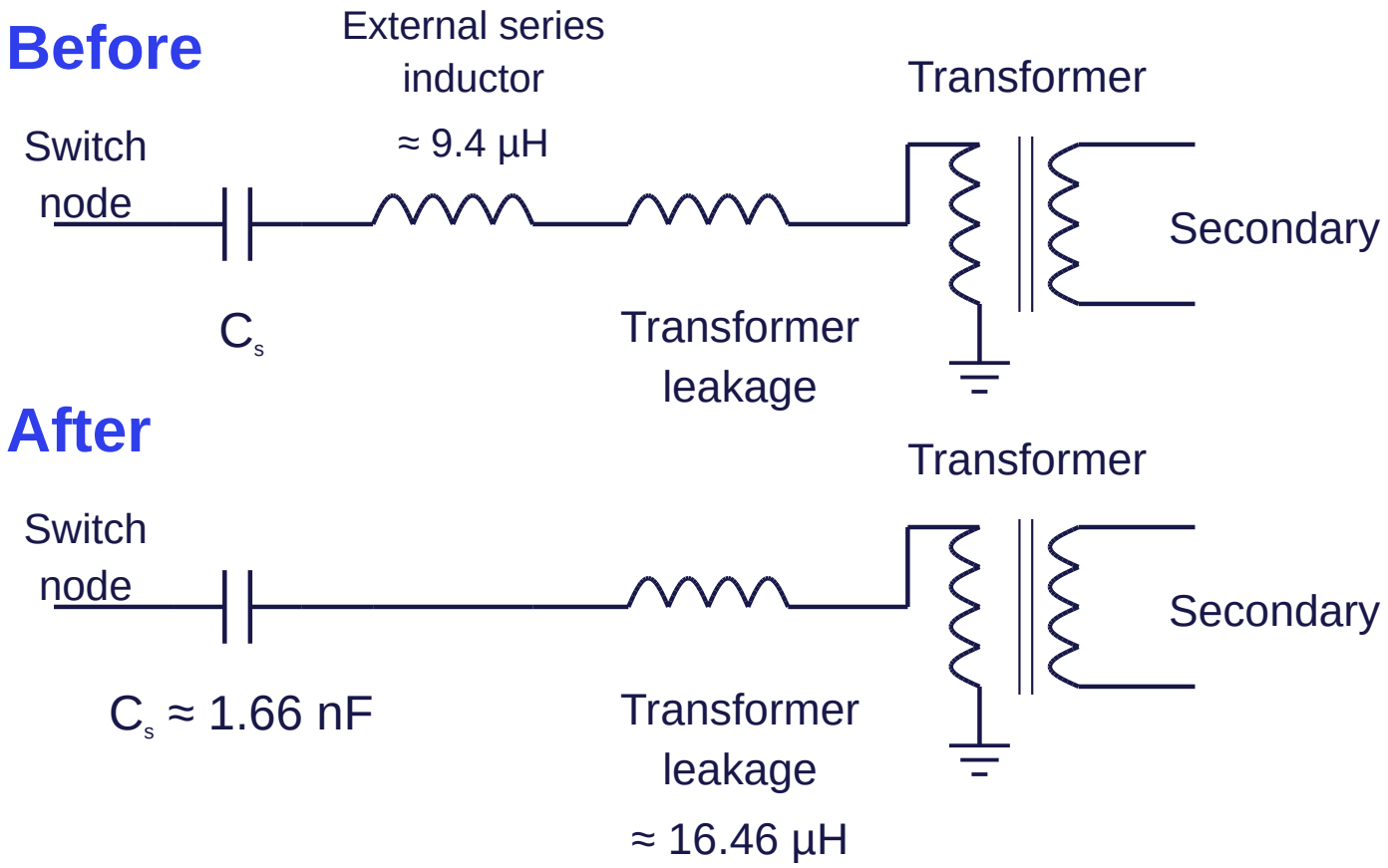


Input current THD



- At 4.0 A: PF 0.99, current THD 3.98%

V2 tank uses transformer leakage alone

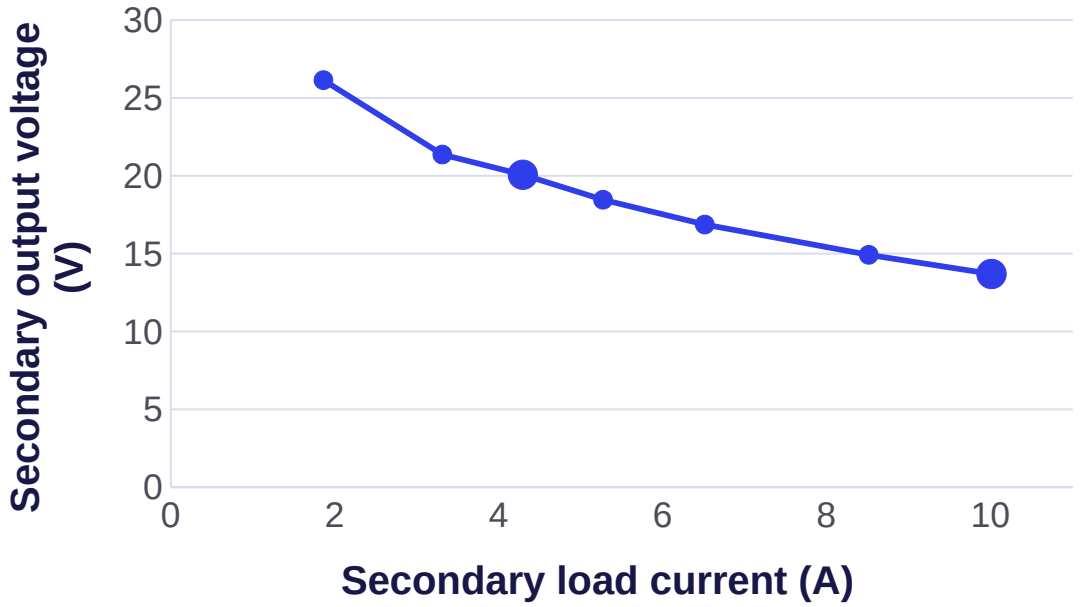


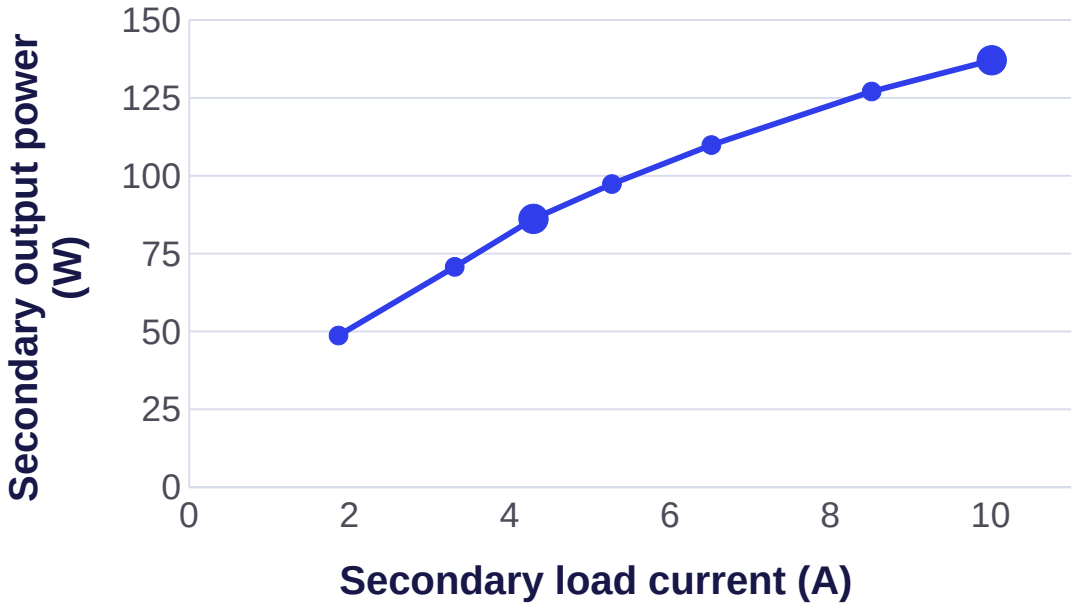
Simplified primary-referred resonant paths

- The hot external series inductor was removed.
- Calculated tank resonance: ≈ 0.963 MHz
- Half-bridge heating fell after tank and timing changes.

V2 secondary power increased as voltage fell

Electronic load at V2 secondary output · ≈240 V AC, 50 Hz





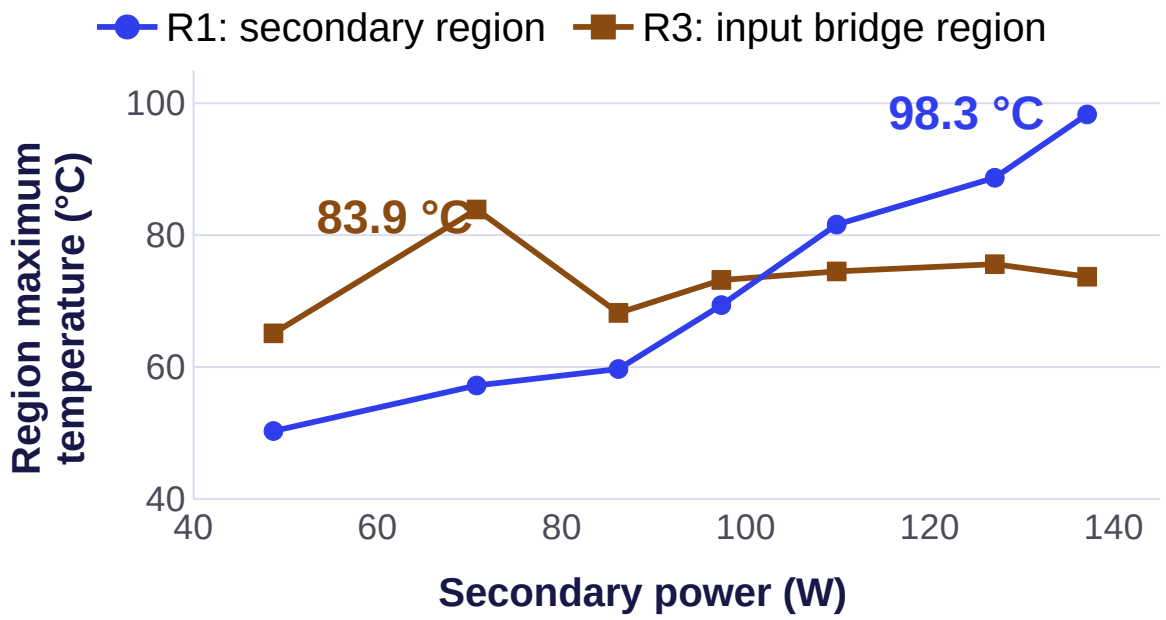
86.17 W at 20.07 V and 4.29 A

137.08 W at 13.69 V and 10.01 A

Higher power came with voltage droop.

V2 secondary region reached 98.3 °C

Thermal-region maxima



Final thermal image at 137.08 W secondary



- No load point reached thermal equilibrium.
- Secondary rectifier current and thermal paths need work.

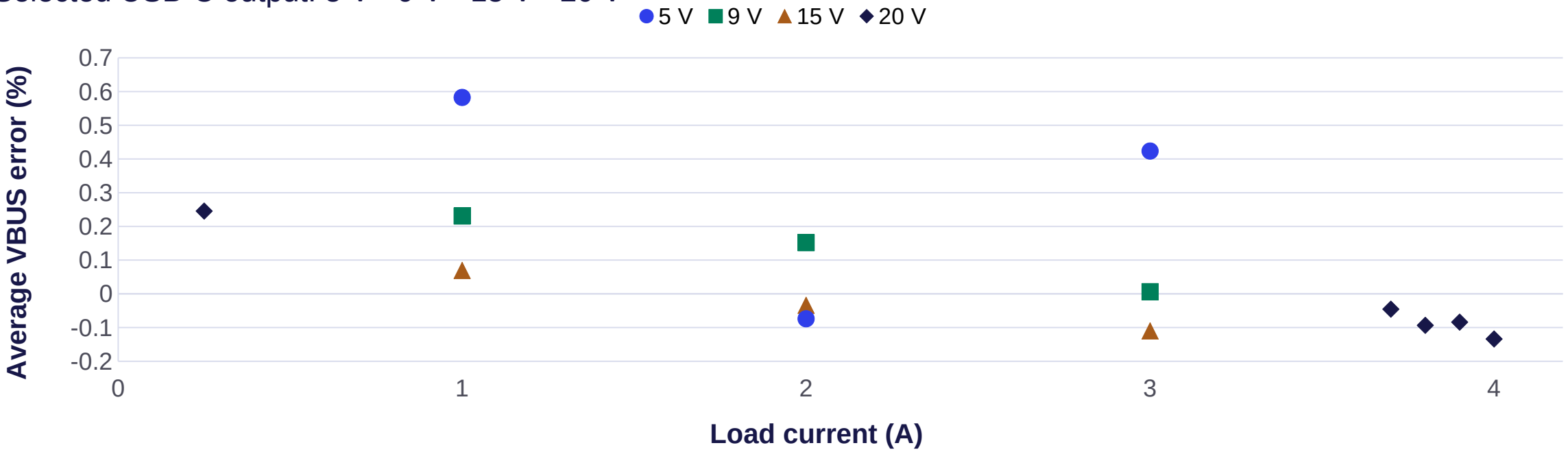
USB-C conversion with digital burst control

- Integrated board, MCU burst control and USB-C PD output.
- V1: 79.89 W at the USB-C connector.
- V2: 137.08 W at secondary, with voltage droop and continued heating.

V1 connector regulation across the load sweep

V1 · USB-C connector · 240 V AC, 50 Hz

Selected USB-C output: 5 V · 9 V · 15 V · 20 V



Largest absolute average-voltage error: 0.583% at 5 V, 1 A

Error = (VBUS,avg − Vset) / Vset × 100%

Efficiency and input quality at selected operating points

V1 · 240 V AC, 50 Hz input

USB-C voltage (V)	Load current (A)	Connector power (W)	Input power (W)	Efficiency estimate (%)	Input PF	Input current THD (%)
4.9963	2.00	9.99	14.78	67.61	0.62	28.37
20.0491	0.25	5.01	8.78	57.09	0.44	79.46
19.9909	3.70	73.97	95.88	77.15	0.97	2.23
19.9813	3.80	75.93	98.99	76.70	0.98	2.62
19.9732	4.00	79.89	104.42	76.51	0.99	3.98

V2 secondary power rose as voltage fell

Electronic load at V2 secondary · approximately 240 V AC, 50 Hz

Secondary voltage (V)	Secondary current (A)	Secondary power (W)	Input bridge R3 region max (°C)	Secondary R1 region max (°C)
26.14	1.86	48.68	65.1	50.3
21.36	3.31	70.75	83.9	57.2
20.07	4.29	86.17	68.2	59.7
18.46	5.27	97.35	73.2	69.4
16.87	6.51	109.87	74.5	81.6
14.92	8.51	127.05	75.6	88.7
13.69	10.01	137.08	73.7	98.3

Thermal equilibrium was not reached.

Tank resonance with and without the external inductor

$$f_0 = 1 / (2\pi\sqrt{(L_{total}\ C_s)})$$

Original design calculation

$L_{total} \approx 9.4 + 16.5 = 25.9\ \mu H$

$C_s = 1\ nF$

$f_0 \approx 0.989\ MHz$

Modified V2 calculation

$L_{total} \approx 16.46\ \mu H$ leakage

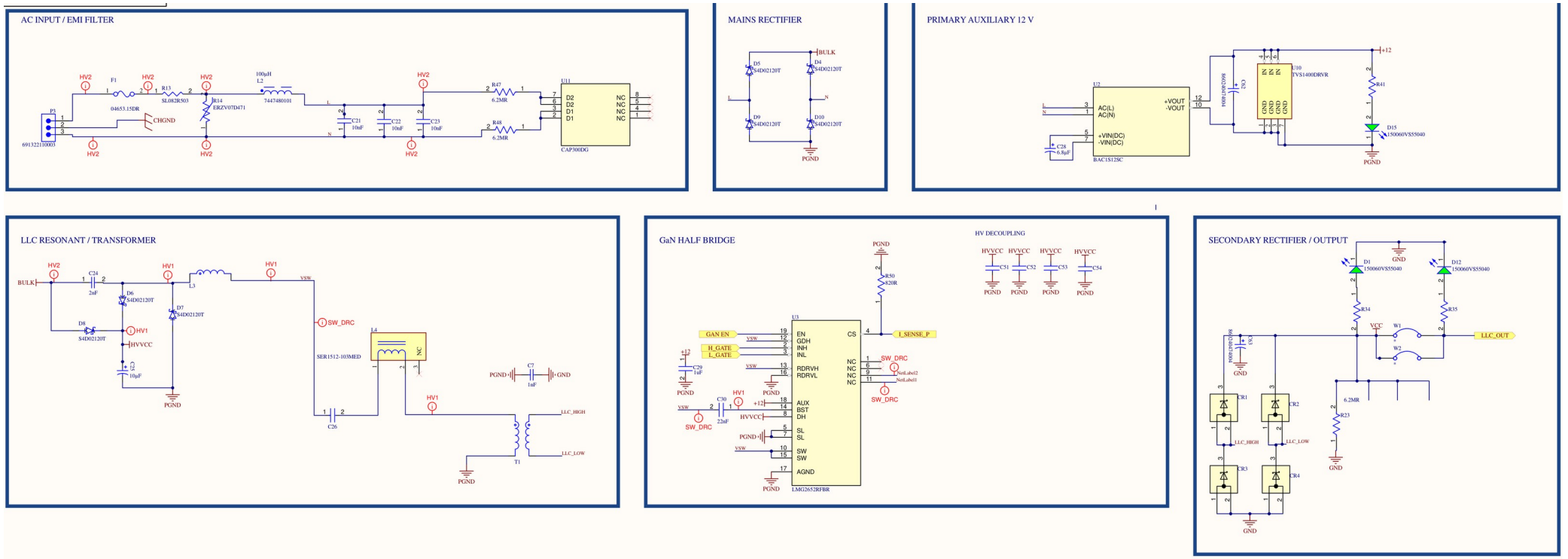
$C_s \approx 1.66\ nF$

$f_0 \approx 0.963\ MHz$

Switching carrier: 1 MHz

Ideal series resonance

V2 power schematic and topology references



V2 power circuit, as drawn

MCU: secondary bulk regulation | USB-C PD: selected connector voltage

Topology: Ammar et al., IEEE JESTPE, 2020, 2021